PRACTICAL: O5

Aim: Do cost and Effort Estimation using Software Cost Estimation Model Objectives: To make use of COCOMO model to find out the cost of software development.

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Organic: A development project can be considered of organic type, if the project deals with developing a well understood application program, the size of the development team is reasonably small, and the team members are experienced in developing similar types of projects.

Model	Project Size	Nature of Project	Innovation	Dead Line	Development Environment
Organic	Typically **2-50 KLOC**	Small Size Project, Experienced developers in the familiar environment, E.g. Payroll, Inventory projects etc.	Little	Not Tight	Familiar & In-house

COCOMO (Constructive Cost Estimation Model) was proposed by **Boehm**
According to Boehm, software cost estimation should be done through three stages:

- Basic COCOMO,
- Intermediate COCOMO, and
- Complete COCOMO

We are going for basic **COCOMO Model**

Basic COCOMO Model: The basic COCOMO model gives an approximate estimate of the project parameters. The basic COCOMO estimation model is given by the following

expressions: **EFFORT** = a1 * (KLOC)^a2 PM
Tdev =b1 x (EFFORT)^b2 Months

- KLOC is the estimated size of the software product expressed in Kilo Lines of Code,
- a_1, a_2, b_1, b_2 are constants for each category of software products,
- Tdev is the estimated time to develop the software, expressed in months,
- Effort is the total effort required to develop the software product, expressed in person months (PMs).
- Every line of source text should be calculated as one LOC irrespective of the actual number of instructions on that line
- If a single instruction spans several lines (say n lines), it is considered to be nLOC.
- The values of a_1, a_2, b_1, b_2 for different categories of products (i.e., organic, semidetached, and embedded) as given by Boehm.

<u>Cost Estimation of Software</u>
Requirements:

External Input =

7 External

Output = 2

No of Inquires – Status, Member information,

Profit/loss No of External Files- Data bases

No of External Interfaces = Client Machine

EI = 7 * 3 = 21

EO = 2 * 4 = 8

EQs = 3 * 3 = 9

IFS = 1 * 7 = 7

EIFS= 2 * 5 = 10

Count Total = 55
Value Adjustment Factors (VAF)
F1: Does the system require reliable backup & recovery? 5

F2: Are data communications
required? 5

F3: Are there distributed processing
functions? 1

F4: Is performance
critical? 3

F5: Will the system run in an existing, heavily utilized operational
environment? 0

F6: Does the system require online
data entry? 2

F7: Does the online data entry require the input transaction to be built over
multiple screens or operation? 3
F8: Are the master files
updated online? 4

F9: Are the inputs, outputs, files or inquiries
complex? 1

F10: Is the internal processing
complex? 1

F11: Is the code designed to be
reusable? 2

F12: Are the conversion and installation included
in design? 1

F13: Is the system design for multiple installations in different
organizations? 2

F14: Is the application designed to facilitate change and ease of use by
the user? 4

$\sum FI=5+5+1+3+2+3+4+1+1+2+1+1+2+4=34$

Function Point = Count Total* [0.65*0.01($\sum$Fi)]

Function Point = 55*[0.65*0.01(34)]

Function Point =

54.45 FP Now 1 FP =

75 LOC in C++

So, for 54.45 FP = 54.45 *75 = 4083.750 LOC

Which is approximately

So, by the Basic COCOMO model the Effort and Development time required for our Organic type Software is

EFFORT = a1 * (KLOC)^a2 PM

=2.4 x (40) ^1.05

=115 PM

Tdev =b1 x (EFFORT)^b2 Months

= 2.5 x (115) ^0.38

= 15 Months

If the salary of the developer is Rs. 10,000/- then the cost of development will be Rs. 1,15,000/-

PRACTICAL: O6

Aim: Prepare System Analysis and System Design of identified Requirement specification using structure design as DFD with data dictionary and Structure chart for the specific module.

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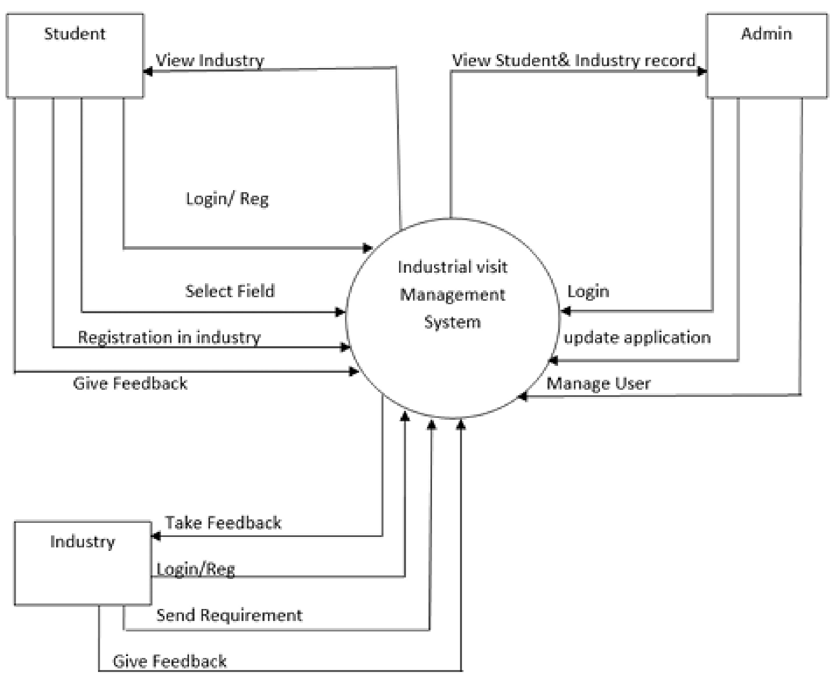
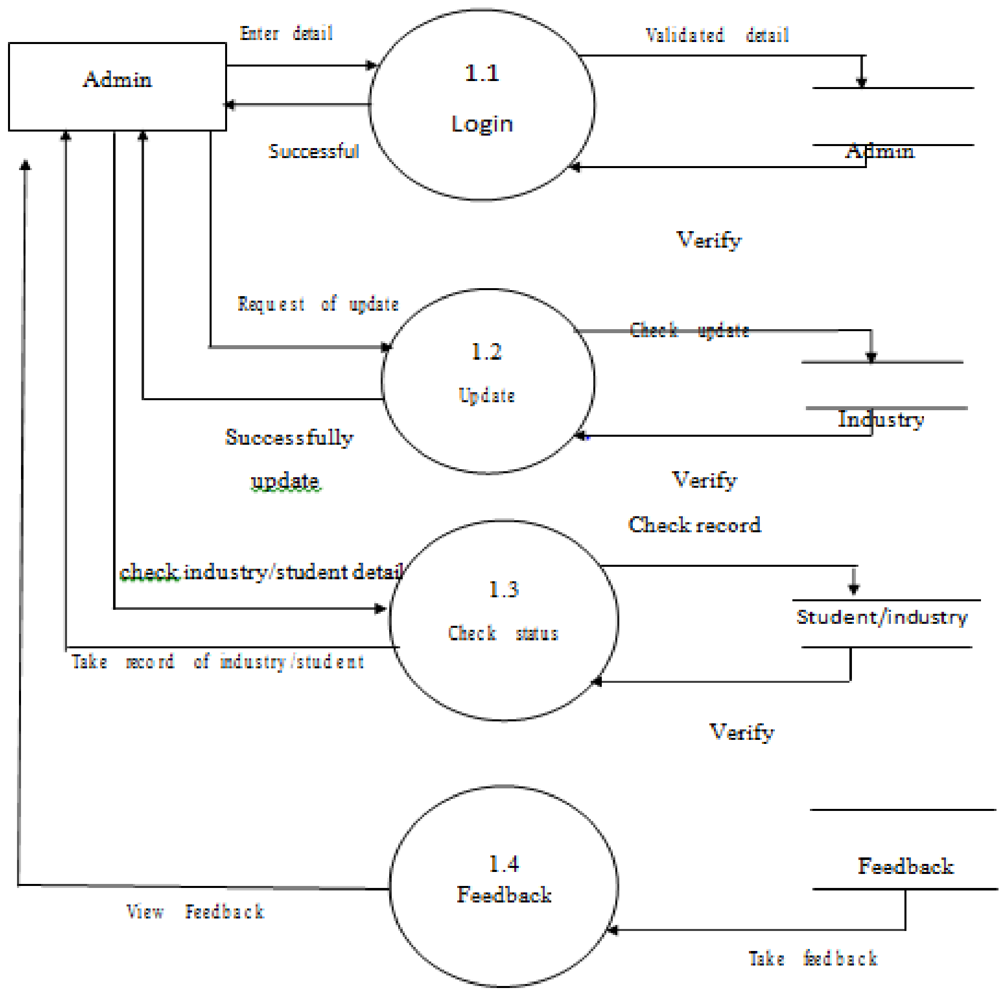
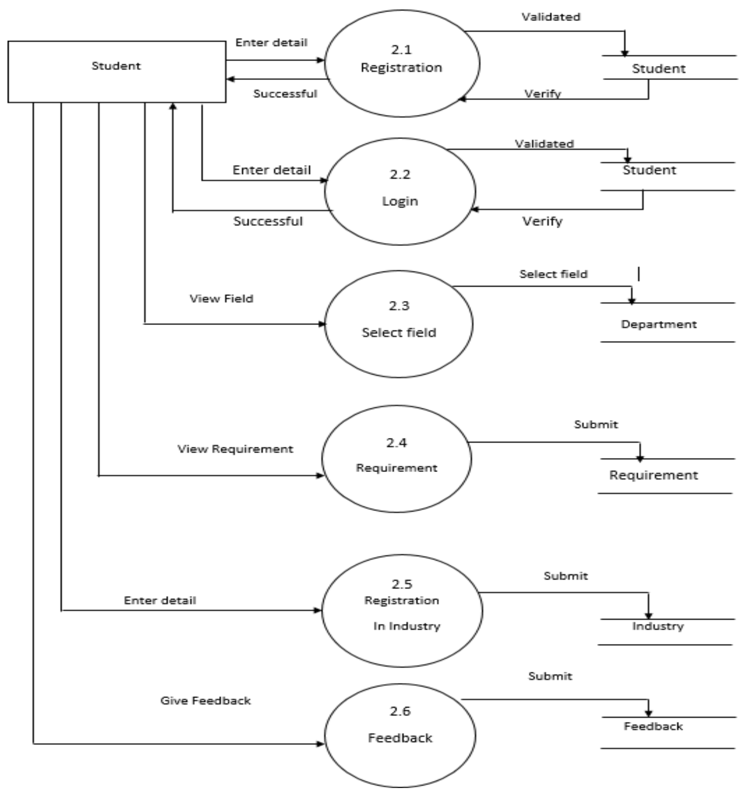
Data Flow Diagram



Admin DFD Diagram



Student DFD Diagram



Industry DFD Diagram